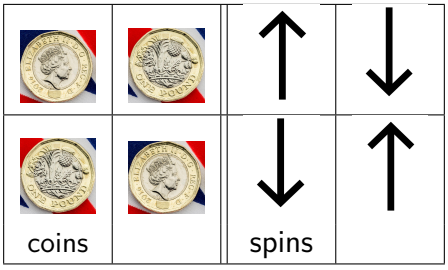
2 A second typical system – a paramagnet

One might ask what the counterpart in coin-tossing is to the the contrast between the time average or the average over phase-space. We discussed a time average in describing one coin tossed N times. Thus N is more the counterpart of T in the time average of observation of the gas. The counterpart to the average over phase space is taking N *identical coins* and counting the number of ways one could have n heads if each coin is tossed *once*. For the coins both processes are clearly equivalent.

The variables E in the gas case and n for coins are usually macroscopic variables, if N is large. We see for each chosen value of those variables there are many configurations of the set of microscopic entities – atoms or coins.

2.1 A dilute magnet – a physical counterpart to coin-tossing

Very dilute spin-$\frac{1}{2}$ atoms in an otherwise insulating and non-magnetic material constitute a physical realisation of the "coins" – the diluteness ensuring they do not interact. They are assumed to be immobile. There are N such atoms, with each atom having two states "up", $\uparrow$, and "down", $\downarrow$. These can be regarded as corresponding to heads and tails.



Then we associate the coin variable $n = n_\uparrow$. Physically the *magnetisation* (this is actually the total magnetic moment, the magnetisation is m/V, but we will be imprecise to avoid long-windedness),

$$m = \mu\,(n_\uparrow - n_\downarrow) \ ,$$

where μ is the magnetic moment associated with the spin (which includes a factor of μ_0 for convenience), is a more useful variable, along with $N = n_\uparrow + n_\downarrow$. We may solve these for $n_\uparrow$ and $n_\downarrow$:

$$n_\uparrow = \frac{1}{2}\left(N + \frac{m}{\mu}\right) \quad \text{and} \quad n_\downarrow = \frac{1}{2}\left(N - \frac{m}{\mu}\right) \ . \tag{2.1}$$

Then if we have an *external* uniform magnetic field, $\mathbf{H} = H\hat{\mathbf{z}}$, up spins have an energy $-\mu H$ (noting the μ_0 has been incorporated in μ) and down spins $+\mu H$, where μ is their magnetic moment. We may then associate an energy with the magnetisation:

$$E = -\mu H\left(n_\uparrow - n_\downarrow\right) = -Hm \ . \tag{2.2}$$

We may use this and Eq. (2.3) to replace m by an expression involving E and H to write $n_\uparrow(N,E)$ and $n_\downarrow(N,E)$:

$$n_\uparrow = \frac{1}{2}\left(N - \frac{E}{\mu H}\right) \quad \text{and} \quad n_\downarrow = \frac{1}{2}\left(N + \frac{E}{\mu H}\right) \ . \tag{2.3}$$

We may now write the number of configurations of $\uparrow$ and $\downarrow$ spins, as a function of E and H:

$$\begin{aligned} W(N,E) = W(N,-Hm) &= W\left(N,-\mu H\left(n_\uparrow - n_\downarrow\right)\right) \\ &= \frac{N!}{n_\uparrow! n_\downarrow!} \\ \text{(Use Eq. (2.3))} \quad &= \frac{N!}{\left[\frac{1}{2}\left(N - \frac{E}{\mu H}\right)\right]!\left[\frac{1}{2}\left(N + \frac{E}{\mu H}\right)\right]!} \ . \end{aligned}$$

Thus using Boltzmann's expression we may deduce the entropy:

$$S(N,E) = k_{\mathrm{B}} \ln W(N,E) \ .$$

This provides a link between the number of *microstates* (where one specifies the state of each spin) which corresponds to a *macrostate* labelled by a macroscopic variable – such as m or E or indeed the entropy, S.

2.2 Electromagnetism and spins in a field – non-examinable



Figure 2.1: Böttcher, who wrote the definitive work on polarization.

The reason for being a little careful about stipulating the field to be "external" is that the full field, $\mathbf{B}$, inside the material reflects the spins contributing to the total field

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) \ , \quad \text{where } \mu_0\mathbf{M} = \frac{m}{V}\hat{\mathbf{z}} \ , \tag{2.4}$$

i.e. magnetisation, m defined above, per unit volume. The external field, $\mathbf{H}$, is caused by current densities, $\mathbf{j}(\mathbf{r})$, external to the system which determine $\mathbf{H}$ through the Maxwell equation:

$$\nabla \times \mathbf{H} = \mathbf{j}(\mathbf{r}) \ .$$

Another difficulty stems from $\mathbf{B}$ in Eq. (2.4) being the **average, macroscopic, field** in the material, which includes the effect of the

spin being considered as well as all of the others. However the spin does not see its own field, which must be subtracted with the result being a "local" field, $\mathbf{B}_{\text{loc}}$, due to all the other spins and the external field. For more detail see the discussion for the equivalent problem in dielectrics in the e-book *Electromagnetism* Grant and Philips p 59-70. And in a more satisfactory manner in Volume 1 of *Theory of electric polarization : Volume 1. Dielectrics in static fields* by C.J.F. Böttcher where there is a lot of detail.

As one can see this is a little complicated. We take refuge in considering a **dilute** set of spins (i.e. most atoms in the materail do not have spins), so that $\mathbf{M} \ll \mathbf{H}$, and to a good approximation we may just use the external field.

3 Temperature and the Canonical Ensemble

Here we will introduce the concept of a heat bath and allow bodies to exchange energy. This allows us to extract temperature as a variable in the case where two bodies are connected and may exchange energy. We then examine the case where one body is much larger than the other and extract the Boltzmann factor exploiting the disparity in scale.

3.1 Connection between number of states and thermodynamic entropy

Here we will provide a simple, but not general, determination of the connection. We start from thermodynamics and then make a plausible association with numbers of configurations.

Consider a *small* isothermal expansion, from V to $V + \Delta V$, of an ideal (classical) gas of N. atoms. The definition of small is $\Delta V/V \ll 1$. This is achieved by heating and allowing the gas to do work to ensure the internal energy remains the same (remember this is an ideal gas, so $U(T)$ is *only* a function of T).

Since the number of states for the gas is plausibly only altered by the change in volume so the number of states when the volume is V, $W(N,V) \propto V^N$ (ie the atoms are all statistically independent). Then:

$$\frac{W(N,V+\Delta V)}{W(N,V)} = \frac{(V+\Delta V)^N}{V^N} = \left(\frac{(V+\Delta V)}{V}\right)^N = \left(1+\frac{\Delta V}{V}\right)^N . \tag{3.1}$$

Now T is constant, so $\Delta U = 0$ for an ideal gas, thus from the First Law of thermodynamics and the equation of state

$$\Delta Q = p\Delta V = Nk_{\text{B}}T\frac{\Delta V}{V} \Rightarrow \frac{\Delta Q}{Nk_{\text{B}}T} = \frac{\Delta V}{V} .$$

We may now rewrite Eq. (3.1) as:

$$\frac{W(N,V+\Delta V)}{W(N,V)} = \left(1+\frac{\Delta Q}{Nk_{\rm B}T}\right)^N$$

$$\text{(Take logs)} \quad \ln W(N,V+\Delta V) - \ln W(N,V) = N\ln\left(1+\frac{\Delta Q}{Nk_{\rm B}T}\right)$$

$$\text{(As } \Delta Q/N \ll 1\text{)} \quad \simeq \frac{\Delta Q}{k_{\rm B}T}$$

$$\Rightarrow k_{\rm B}\left(\ln W(N,V+\Delta V) - \ln W(N,V)\right) = \frac{\Delta Q}{T}$$

$$\Delta\left[k_{\rm B}\ln W(N,V)\right] = \frac{\Delta Q}{T}\ .$$

Now compare the last equality with the thermodynamic result

$$T\Delta S = \Delta Q \quad \Rightarrow \quad \Delta S = \frac{\Delta Q}{T}\ ,$$

to identify

$$S = k_{\rm B}\ln W(N,V)\ ,$$

i.e. Boltzmann's result

$$S = k_{\rm B}\ln W\ .$$

This links the macroscopic (thermodynamic) and statistical (microscopic) description, in this special case. **We will boldly apply this result below to cases where the number of states is not only labelled by the volume but by the energy as well.**

3.2 Probabilities

Consider one system, where the number of configurations with energy E is $W(E)$. We assume each of the configurations is equally likely (eg due to the even coverage of phase space in the ergodic hypothesis), and this is termed *equal a priori probability*.

Thus the probability, p, of *one* configuration is

$$p = \frac{1}{W(E)}\ .$$

Notice these configurations are *mutually exclusive* – you cannot be observing more than one simultaneously – so this probability distribution is normalised if we add all the possible outcomes:

$$\underbrace{\frac{1}{W(E)} + \frac{1}{W(E)} + \cdots + \frac{1}{W(E)}}_{W(E)\ \text{terms}} = 1\ .$$

A little terminology:

- Each configuration is a *microstate*, where the complete microscopic information labels them (eg the positions and momenta of all of the particles in the gas, or *which* spins are "up" or "down").
- A *macrostate*, in this case, is labelled by E, but with no other information about the constituents.

3.3 Two bodies in thermal contact

Let us now consider two bodies capable of exchanging energy with the number of states at energy E_1 and E_2 being $W_1(E_1)$ and $W_2(E_2)$. We assume the volume of each body is fixed, and the two bodies are isolated from the "rest of the Universe", so the total energy, $E = E_1 + E_2$, is fixed.

Let us assume the coupling between the bodies (to allow transport of energy) is non-zero but very small. Then we may deduce the total number of states by considering the systems separately. Using the relation Eq. (3.1) between W and entropy, S, will allow us to deduce the temperature of the two connected bodies.

Then the total number of states where $E_1 + E_2 = E$ is

$$W_{\text{tot}}(E, E_1) = W_1(E_1)W_2(E - E_1) \ .$$

The most probable value for E_1 is where

$$\begin{aligned}
& \frac{\partial W_{\text{tot}}}{\partial E_1} = 0 \\
\Rightarrow \quad & W_1'(E_1)W_2(E - E_1) - W_1(E_1)W_2'(E - E_1) = 0 \\
& \Rightarrow \quad \frac{W_1'(E_1)}{W_1(E_1)} = \frac{W_2'(E - E_1)}{W_2(E - E_1)} \\
\text{(Note } E - E_1 = E_2\text{.)} \quad & \Rightarrow \quad k_{\text{B}} \frac{\mathrm{d} \ln W_1(E_1)}{\mathrm{d}E_1} = k_{\text{B}} \frac{\mathrm{d} \ln W_2(E_2)}{\mathrm{d}E_2} \\
& \Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}E_1} \left[k_{\text{B}} \ln W_1(E_1) \right] = \frac{\mathrm{d}}{\mathrm{d}E_2} \left[k_{\text{B}} \ln W_2(E_2) \right] \ . \qquad (3.2)
\end{aligned}$$

Now remember our identification of $S = k_{\text{B}} \ln W$ and use

$$\left(\frac{\partial S}{\partial E} \right)_V = \frac{1}{T} \ ,$$

to deduce from Eq. (3.2), noting the system's volumes are constant:

$$\begin{aligned}
& \left(\frac{\partial S_1}{\partial E_1} \right)_{V_1} = \left(\frac{\partial S_2}{\partial E_2} \right)_{V_2} \\
\Rightarrow \quad & \frac{1}{T_1} = \frac{1}{T_2} \ .
\end{aligned}$$

So the temperatures are equal in the two bodies at the most probable values of E_1 and E_2, i.e. they are in thermal equilibrium.

Aside:
This discussion comes from regarding the entropy as a function of $E = U$ and V, then

$$\begin{aligned} \mathrm{d}U &= T\mathrm{d}S - p\mathrm{d}V \\ \Rightarrow \quad \frac{1}{T}\mathrm{d}U &= \left(\frac{\partial S}{\partial V}\right)_U \mathrm{d}V + \left(\frac{\partial S}{\partial U}\right)_V \mathrm{d}U - p\mathrm{d}V \\ \text{(Constant } V\text{, so } \mathrm{d}V = 0.) \quad \frac{1}{T}\mathrm{d}U &= \left(\frac{\partial S}{\partial U}\right)_V \mathrm{d}U \\ \Rightarrow \quad \frac{1}{T} &= \left(\frac{\partial S}{\partial U}\right)_V . \end{aligned}$$

To see that the *overall* temperature, $T_{(1 \text{ and } 2)}$, of the assembly of the two systems together is the same as the individual temperatures, $T_1 = T_2$, is a bit more tricky – see non-examinable Appendix A. If you think this is obvious, that is fine!

To see the zeroth law of thermodynamics emerge from a similar approach, see appendix B.

3.4 System and "Heat bath"

Let us now consider the case where we have a "small" system able to exchange energy with a very large system, called a *heat bath* or *thermal reservoir*. Any change of energy between them is assumed to be very small compared to the total energy of the bath.

Then if $W_\mathsf{s}(E_\mathsf{s})$ is the number of configurations of the system with energy E_s and $W_\mathsf{b}(E_\mathsf{tot} - E_\mathsf{s})$ is the number of configurations of the bath, with energy $E_\mathsf{b} = E_\mathsf{tot} - E_\mathsf{s}$, we may write the total number of configurations as:

$$W_\mathsf{tot}(E_\mathsf{tot}, E_\mathsf{s}) = W_\mathsf{s}(E_\mathsf{s})\, W_\mathsf{b}(E_\mathsf{tot} - E_\mathsf{s}) \ .$$

where we assume the means of energy exchange does not significantly alter the counting of the configurations – so that we may assume the two bodies are statistically independent.[1]

Now we will exploit $E_\mathsf{s} \ll E_\mathsf{tot}$ to Taylor expand W_b. However it is much more convenient to proceed in a slightly roundabout way – as we understand derivatives of $S = k_\mathrm{B} \ln W$ more than derivatives of W itself.

[1] One might worry if one makes the means of energy exchange very weak, then maybe the energies would be conserved in both bodies separately. But we may wait for as long as we want and the coupling will make itself felt if one waits long enough.

$$
\begin{aligned}
W_{\text{tot}}\left(E_{\text{tot}}, E_{\text{s}}\right) &= W_{\text{s}}(E_{\text{s}})\, W_{\text{b}}(E_{\text{tot}} - E_{\text{s}}) \\
&= W_{\text{s}}(E_{\text{s}})\, \exp\left[\ln W_{\text{b}}(E_{\text{tot}} - E_{\text{s}})\right] \\
\left[\text{Note } \frac{\partial E_{\text{b}}}{\partial E_{\text{s}}} = -1\right] \quad &= W_{\text{s}}(E_{\text{s}})\, \exp\left[\ln W_{\text{b}}(E_{\text{tot}}) - E_{\text{s}} \left.\frac{\partial \ln W_{\text{b}}}{\partial E_{\text{b}}}\right|_{E_{\text{b}}=E_{\text{tot}}} + \cdots\right] \\
&\simeq W_{\text{s}}(E_{\text{s}})\, W_{\text{b}}(E_{\text{tot}})\, \exp\left[-\left(\frac{\partial S_{\text{b}}}{\partial E_{\text{b}}}\right)_{V_{\text{b}}} \frac{E_{\text{s}}}{k_{\text{B}}}\right] \\
\left[\text{Use } \left(\frac{\partial S_{\text{b}}}{\partial E_{\text{b}}}\right)_{V_{\text{b}}} = \frac{1}{T_{\text{b}}}\right] \quad &= W_{\text{s}}(E_{\text{s}})\, W_{\text{b}}(E_{\text{tot}}) \exp\left(-\frac{E_{\text{s}}}{k_{\text{B}} T_{\text{b}}}\right) \ . \qquad (3.3)
\end{aligned}
$$

Now add the contributions from each value of E_{s}, and since a continuous variable this is an integral:

$$
\begin{aligned}
W(E_{\text{tot}}) &= \int \mathrm{d}E_{\text{s}}\; W_{\text{tot}}\left(E_{\text{tot}}, E_{\text{s}}\right) \simeq W_{\text{b}}(E_{\text{tot}}) \int \mathrm{d}E_{\text{s}}\; W_{\text{s}}(E_{\text{s}}) \exp\left(-\frac{E_{\text{s}}}{k_{\text{B}} T_{\text{b}}}\right) \\
\Rightarrow \quad W_{\text{s}}(T, V) &= \frac{W(E_{\text{tot}})}{W_{\text{b}}(E_{\text{tot}})} \simeq \int \mathrm{d}E_{\text{s}}\; W_{\text{s}}(E_{\text{s}}) \exp\left(-\frac{E_{\text{s}}}{k_{\text{B}} T_{\text{b}}}\right) = Z_{\text{s}}(T, V) \ . \qquad (3.4)
\end{aligned}
$$

where we have defined the conventional symbol, $Z_{\text{s}}(T, V)$, the *canonical partition function*, for the system when in contact with a heat bath at temperature T.

An application of the approach of this subsection and the last to two systems which may exchange *volume* as well as energy is discussed in problems 1.7 and 1.8 (the latter examining a "volume" bath).

The significance of this can be illustrated by thinking of the probability distribution, $p(E_{\text{s}}|E_{\text{tot}})$, for the observation of the system having energy E_{s}, for a fixed value of E_{tot}. This is clearly proportional to the number of configurations, $W_{\text{tot}}(E_{\text{tot}}, E_{\text{s}})$, and one may readily check that

the normalised (conditional) probability distribution is:

$$
\begin{aligned}
p(E_{\mathsf{s}}|E_{\mathsf{tot}}) &= \frac{W_{\mathsf{tot}}(E_{\mathsf{tot}}, E_{\mathsf{s}})}{\int \mathrm{d}E_{\mathsf{s}}\; W_{\mathsf{tot}}(E_{\mathsf{tot}}, E_{\mathsf{s}})} \\
\text{(Use Eq. (3.3))} \quad &\simeq \frac{W_{\mathsf{b}}(E_{\mathsf{tot}}) W_{\mathsf{s}}(E_{\mathsf{s}})\; \exp\left(-\dfrac{E_{\mathsf{s}}}{k_{\mathrm{B}}T_{\mathsf{b}}}\right)}{\int \mathrm{d}E_{\mathsf{s}}\; W_{\mathsf{b}}(E_{\mathsf{tot}}) W_{\mathsf{s}}(E_{\mathsf{s}})\; \exp\left(-\dfrac{E_{\mathsf{s}}}{k_{\mathrm{B}}T_{\mathsf{b}}}\right)} \\
\text{(Note } W_{\mathsf{b}} \text{ constant.)} \quad &= \frac{W_{\mathsf{b}}(E_{\mathsf{tot}})}{W_{\mathsf{b}}(E_{\mathsf{tot}})} \frac{W_{\mathsf{s}}(E_{\mathsf{s}})\; \exp\left(-\dfrac{E_{\mathsf{s}}}{k_{\mathrm{B}}T_{\mathsf{b}}}\right)}{\int \mathrm{d}E_{\mathsf{s}}\; W_{\mathsf{s}}(E_{\mathsf{s}})\; \exp\left(-\dfrac{E_{\mathsf{s}}}{k_{\mathrm{B}}T_{\mathsf{b}}}\right)} \\
&= \frac{W_{\mathsf{s}}(E_{\mathsf{s}})\; \exp\left(-\dfrac{E_{\mathsf{s}}}{k_{\mathrm{B}}T_{\mathsf{b}}}\right)}{\int \mathrm{d}E_{\mathsf{s}}\; W_{\mathsf{s}}(E_{\mathsf{s}})\; \exp\left(-\dfrac{E_{\mathsf{s}}}{k_{\mathrm{B}}T_{\mathsf{b}}}\right)} \\
&= \frac{W_{\mathsf{s}}(E_{\mathsf{s}})\; \exp\left(-\dfrac{E_{\mathsf{s}}}{k_{\mathrm{B}}T_{\mathsf{b}}}\right)}{Z_{\mathsf{s}}(T_{\mathsf{b}}, V)} \; .
\end{aligned}
$$

Thus the partition function, $Z_{\mathsf{s}}(T_{\mathsf{b}}, V)$, may be interpreted as the normalisation for $p(E_{\mathsf{s}}|E_{\mathsf{tot}})$. It has a much more profound physical interpretation as we will see in section 4, in its relation to the Helmholtz free energy.

3.5 Canonical partition function for a gas

Let us make contact with the example of a classical (not necessarily ideal) gas with N particles. There we had for the total number of configurations

$$
\begin{aligned}
W_{\mathsf{s}}(E_{\mathsf{s}}, V) &= \frac{1}{\mathsf{h}^{3N}} \int \prod_{n=1}^{N} \mathrm{d}\mathbf{q}_n \mathrm{d}\mathbf{p}_n\; \delta\left(E_{\mathsf{s}} - \mathcal{H}_{\mathsf{s}}(\mathbf{q}_1, \cdots, \mathbf{q}_N; \mathbf{p}_1, \cdots, \mathbf{p}_N)\right) \\
&= \frac{1}{\mathsf{h}^{3N}} \int \mathrm{d}Q \mathrm{d}P\; \delta(E_{\mathsf{s}} - \mathcal{H}_{\mathsf{s}}(Q, P)) \; ,
\end{aligned}
$$

using the shorthand Q and P to denote all the positions and momenta.

Then we see that upon substitution into Eq. (3.4)

$$
\begin{aligned}
& Z_{\mathsf{s}}(T,V) \simeq \int \mathrm{d}E_{\mathsf{s}}\; W_{\mathsf{s}}(E_{\mathsf{s}}) \exp\left(-\frac{E_{\mathsf{s}}}{k_{\mathrm{B}}T_{\mathsf{b}}}\right) \\
\Rightarrow \quad & Z_{\mathsf{s}}(T,V) \simeq \int \mathrm{d}E_{\mathsf{s}}\; \underbrace{\frac{1}{\mathsf{h}^{3N}} \int \mathrm{d}Q\mathrm{d}P\; \delta\left(E_{\mathsf{s}} - \mathcal{H}_{\mathsf{s}}(Q,P)\right)}_{W_{\mathsf{s}}(E_{\mathsf{s}})} \exp\left(-\frac{E_{\mathsf{s}}}{k_{\mathrm{B}}T_{\mathsf{b}}}\right)
\end{aligned}
$$

(Perform integral over E_{s}.)

$$
Z_{\mathsf{s}}(T,V) = \frac{1}{\mathsf{h}^{3N}} \int \mathrm{d}Q\mathrm{d}P \exp\left[-\frac{\mathcal{H}_{\mathsf{s}}(Q,P)}{k_{\mathrm{B}}T_{\mathsf{b}}}\right] \tag{3.5}
$$

So the partition function in Eq. (3.5) is just the integral of the Boltzmann factor for the energy (Hamiltonian) for the gas integrated over all phase space.

3.6 Example: single particle experiencing a potential

The particle has mass m, resides in one dimension and experiences a potential $V(q)$, then the partition function is

$$
Z(T) = \frac{1}{\mathsf{h}} \int_{-\infty}^{\infty} \mathrm{d}q \int_{-\infty}^{\infty} \mathrm{d}p\; \exp\left[-\beta\left(\frac{p^2}{2m} + V(q)\right)\right] \; .
$$

Notice we may use $\mathrm{e}^{a+b} = \mathrm{e}^{a}\mathrm{e}^{b}$ to separate the terms involving momentum, p, (the kinetic energy) and position, q, (the potential energy) in the integrand of the partition function and then separate the integrals:

$$
\begin{aligned}
Z(T) &= \frac{1}{\mathsf{h}} \int_{-\infty}^{\infty} \mathrm{d}q \int_{-\infty}^{\infty} \mathrm{d}p\; \exp\left[-\beta\left(\frac{p^2}{2m}\right)\right]\; \exp\left[-\beta\left(V(q)\right)\right] \\
&= \frac{1}{\mathsf{h}} \int_{-\infty}^{\infty} \mathrm{d}q\; \exp\left[-\beta\left(V(q)\right)\right]\; \int_{-\infty}^{\infty} \mathrm{d}p\; \exp\left[-\beta\left(\frac{p^2}{2m}\right)\right] \\
&= \frac{1}{\mathsf{h}} Z_q(T)\, Z_p(T) \; .
\end{aligned} \tag{3.6}
$$

And the second, momentum, integral can readily be performed as it is a Gaussian integral, $\int_{-\infty}^{\infty} \mathrm{d}x\; \mathrm{e}^{-\alpha x^2} = \sqrt{\pi/\alpha}$, with $\alpha = (2mk_{\mathrm{B}}T)^{-1}$. So

$$
Z(T) = \frac{1}{\mathsf{h}} \int_{-\infty}^{\infty} \mathrm{d}q\; \exp\left[-\beta\left(V(q)\right)\right]\; \sqrt{2\pi m k_{\mathrm{B}}T} \; .
$$

It is hard to overestimate the importance of this factorisation for a classical non-relativistic particle system. The kinetic energy contribution is always of the same form – only the mass of the particle differs from case to case. So any specific or interesting aspects always come from the potential energy component. Sometimes quantum systems may be in a classical regime at high temperatures. In that case the kinetic energy term may be different for condensed matter problems – eg pseudo-relativistic for graphene, and quartic, or worse, functions for the momentum dependence of the kinetic energy.

In the particular case of the harmonic oscillator, $V(q) = \frac{1}{2}Kq^2$, we may use the Gaussian integral again

$$\begin{aligned} Z_{\text{SHO}}(T) &= \frac{1}{\mathsf{h}}\int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left[-\beta\left(\tfrac{1}{2}Kq^2\right)\right] \ \sqrt{2\pi m k_{\text{B}}T} \\ &= \frac{1}{\mathsf{h}}\sqrt{\frac{2\pi k_{\text{B}}T}{K}} \ \sqrt{2\pi m k_{\text{B}}T} \\ &= \frac{1}{\mathsf{h}} 2\pi k_{\text{B}}T\sqrt{\frac{m}{K}} \\ \Rightarrow \quad Z_{\text{SHO}}(T) &= \frac{1}{\mathsf{h}} 2\pi \frac{k_{\text{B}}T}{\omega_0} \\ &= \frac{k_{\text{B}}T}{\hbar\omega_0} \ , \end{aligned} \tag{3.7}$$

where $\omega_0 = \sqrt{K/m}$ is the natural frequency of the oscillator. We will return to the striking simplicity of this result presently.

It is very important to note that this factorisation of the kinetic and potential contributions to Z for a single particle does not carry over to quantum mechanical systems. This is because $\mathrm{e}^{\hat{a}+\hat{b}} \neq \mathrm{e}^{\hat{a}}\mathrm{e}^{\hat{b}}$ in general for *operators* $\hat{a}$ and $\hat{b}$, if $[\hat{a},\hat{b}] \neq 0$. In the quantum counterpart to the case above the operators corresponding to kinetic and potential energy do not commute.

3.7 Averages of quantities from the partition function

With a probability distribution for discrete, p_n, or continuous, $p(x)$, variables, we may construct averages of the variables n and x respectively

$$\langle n\rangle = \sum_{n=1}^{N} n p_n \quad \text{and} \quad \langle x\rangle = \int_a^b \mathrm{d}x \ x p(x) \ ,$$

where $1 \leq n \leq N$ and $a \leq x \leq b$. This may be generalised to functions of n and x:

$$\langle f_n\rangle = \sum_{n=1}^{N} f_n p_n \quad \text{and} \quad \langle f(x)\rangle = \int_a^b \mathrm{d}x \ f(x)p(x) \ .$$

Given the partition function and Boltzmann factors provide a probability for a state of energy E, we have the probability of observing that state being

$$p(E) = \frac{\mathrm{e}^{-\beta E}}{Z} \quad \text{and} \quad \langle E\rangle = \int \mathrm{d}E \ E\,W(E)p(E) = \int \mathrm{d}E \ E\,W(E)\frac{\mathrm{e}^{-\beta E}}{Z}$$

being the average energy, $\langle E\rangle$.

Restricting ourselves to one particle in one dimension for simplicity, we can also determine averages of functions of the momenta of the particle, $f(p)$, or their position, $g(q)$, as, using the factorisation in Eq. (3.6), with a phase-space probability, $p(q,p)$

$$\langle f(p)\rangle = \frac{1}{\mathsf{h}}\int \mathrm{d}p \int \mathrm{d}q\ f(p)\underbrace{\frac{1}{Z}\exp\left[-\beta\left(\frac{p^2}{2m}+V(q)\right)\right]}_{=p(q,p)}$$

$$\text{(Use Eq. (3.6).)} \quad = \frac{1}{\mathsf{h}}\frac{1}{Z_q}\underbrace{\int \mathrm{d}q \exp\left(-\beta V(q)\right)}_{=Z_q}\mathsf{h}\frac{1}{Z_p}\int \mathrm{d}p\ f(p)\exp\left[-\beta\left(\frac{p^2}{2m}\right)\right]$$

$$= \frac{1}{Z_p}\int \mathrm{d}p\ f(p)\exp\left[-\beta\left(\frac{p^2}{2m}\right)\right]\ .$$

Similarly (again noting cancellation of factors of h),

$$\langle g(q)\rangle = \frac{1}{\mathsf{h}}\int \mathrm{d}p \int \mathrm{d}q\ g(q)\underbrace{\frac{1}{Z}\exp\left[-\beta\left(\frac{p^2}{2m}+V(q)\right)\right]}_{=p(q,p)}$$

$$= \frac{1}{Z_q}\int \mathrm{d}q\ g(q)\exp\left(-\beta V(q)\right)\ \frac{1}{Z_p}\underbrace{\int \mathrm{d}p\ \exp\left[-\beta\left(\frac{p^2}{2m}\right)\right]}_{=Z_p}$$

$$= \frac{1}{Z_q}\int \mathrm{d}q\ g(q)\exp\left(-\beta V(q)\right)\ .$$

So the factorisation of the partition function simplifies considerably the calculation of averages of functions of q and p. This utility extends to a single particle in three dimensions ($Z = Z_{\mathbf{q}}Z_{\mathbf{p}}/\mathsf{h}^3$) and to many particles ($Z = Z_Q Z_P/\mathsf{h}^{3N}$) straightforwardly, in general allowing for inter-particle interactions, $U\left(|\mathbf{q}_n - \mathbf{q}_m|\right)$, being included in $Z_{\mathbf{Q}}$ as well as the single particle potential, $V(\mathbf{q}_n)$.

3.8 The nature of the partition integral

Focus on the integral over E_s for the system. A consequence is that (for a gas, for example) the total number of configurations, $W_\mathsf{s}(E_\mathsf{s})$, grows very rapidly with energy, as there are more and more ways to divide up the kinetic energy of the gas between the particles. In contrast, the exponential will drive the integrand to zero at large E_s. There must be (at least one!) a maximum in between the lowest value of E_s and $E_\mathsf{s} = \infty$.

In a gas of N particles residing in d dimensions, for example, the total number of configurations grows approximately as $E^{dN/2}$ (see Eqn (B.2) from Appendix B in section 1). The Boltzmann factor decays exponentially with E, $\exp(-\beta E)$ ($\beta = 1/k_\mathrm{B}T$), motivating the following simple representation of this type of functions.

Consider the function $f(x) = x^n \mathrm{e}^{-bx}$, then clearly there is a maximum, denoted by x^*. Where is it?

$$\begin{aligned} f'(x^*) &= (n{x^*}^{(n-1)} - b{x^*}^n)\mathrm{e}^{-bx^*} = 0 \\ \Rightarrow \text{either an } (n-1)\text{-fold zero at } x^* = 0 \text{ or } \quad & n - bx^* = 0 \\ \Rightarrow \quad & x^* = \frac{n}{b} \text{ or } x^* = 0 \ . \end{aligned}$$

Now the second derivative – to see how narrow the peak is:

$$\begin{aligned} f''(x) &= \left[n(n-1)(x^*)^{n-2} - 2bn(x^*)^{n-1} + b^2(x^*)^n\right] \mathrm{e}^{-bx^*} \\ \text{(Substitute } x^* = n/b.) \quad f''(x^*) &= \left(\frac{n}{b}\right)^{n-2} \mathrm{e}^{-n} \left[n(n-1) - 2n^2 + n^2\right] \\ &= -n \left(\frac{n}{b}\right)^{n-2} \mathrm{e}^{-n} \ . \end{aligned}$$

Since the factor of n^n will dominate, $f''(x^*)$ is negative and becomes large as $n \to \infty$ – a sharp maximum.

This approach is made more systematic in problem 1.3 on the derivation of Stirling's approximation. And is also the basis of the approximation to derive classical thermodynamics from statistical mechanics.

We will see the physical interpretation, in terms of thermodynamics, of this maximum in the next section.

3.9 Summary

In this section we considered a system which could exchange energy with its environment. We showed:

- the temperature of two coupled systems is the same,
- the Boltzmann factor emerges from considering a heat bath (system with very large thermal capacity) coupled to the system of interest,
- how to construct averages over energy,
- how to construct averages over functions of q and p,
- stressed the separation of integration over position and momentum variables in performing averages for *classical* systems,
- but noted this was not true for quantum systems.

Appendices

A Microcanonical temperature of two-systems (non-examinable)

We have two thermally coupled systems with numbers of states $W_1(E_1)$ and $W_2(E_2)$. The total number of states, $W(E)$, when the combined system is coupled (weakly) with total energy E (then $E_2 = E - E_1$) is

$$W(E) = \int \mathrm{d}E_1 \; W_1(E_1) W_2(E - E_1) \; . \tag{A.1}$$

We believe that the product of the two W's is sharply peaked as the number, N, of constituent objects (particles, spins etc) becomes large.

There is a technique for systematically approximating integrals with sharp peaks called *Laplace's method*. Write

$$I = \int \mathrm{d}x \; f(x) = \int \mathrm{d}x \; \mathrm{e}^{\ln f(x)} \; .$$

We will assume the only stationary point of the argument of the exponential is real and within the region of integration, say x^*.

$$\frac{\mathrm{d}}{\mathrm{d}x} \ln f(x) = \frac{f'(x)}{f(x)} = 0 \quad \text{if } f'(x^*) = 0 \; .$$

Then we Taylor expand $\ln f(x)$ around x^*, noting there is no linear term as x^* is a stationary point, and suppressing the fact that $(\ln f)''$ is evaluated at $x = x^*$ we see :

$$I = \int \mathrm{d}x \; \exp\left\{ \ln[f(x = x^*)] + \frac{(\ln f)''}{2}(x - x^*)^2 + \cdots \right\} \; .$$

We now discard the other terms (see chapter 2 of the e-book *Asymptotic analysis* by J. D. Murray.). Then, noting that $(\ln f)'' < 0$ for maximum:

$$\begin{aligned} I &\simeq f(x^*) \int \mathrm{d}x \; \exp\left\{ -\frac{|(\ln f)''|}{2}(x - x^*)^2 \right\} \\ &\simeq f(x^*) \sqrt{\frac{\pi}{|(\ln f)''|}} \; . \end{aligned} \tag{A.2}$$

Here we have assumed that we may extend the region of integration (where the integrand is exponentially small) to the whole real axis and perform the Gaussian integral. The errors are in general very small compared to the corrections from the rest of the Taylor series of the argument of the exponential.

Let us apply Eq. (A.2) to Eq. (A.1).

$$
\begin{aligned}
& f(E, E_1) = \ln\{W_1(E_1) W_2(E - E_1)\} \\
\Rightarrow \quad & \frac{\partial f}{\partial E_1} = \frac{W_1'}{W_1} - \frac{W_2'}{W_2} = 0 \\
\Rightarrow \quad & \frac{1}{T_1^{\text{can}}} = \frac{1}{T_2^{\text{can}}} \, .
\end{aligned} \tag{A.3}
$$

Where we have recognized the appearance of equal canonical temperatures of the two subsystems.

Let the stationary point occur at $E_1 = E_1^*(E)$, i.e. $E_1^*(E)$ is the solution of the equation above. Then the approximate value for $W(E)$ is:

$$
W(E) \simeq \sqrt{\frac{\pi}{[\ln W_1(E_1) + \ln W_2(E - E_1)]''|_{E_1 = E_1^*(E)}}} W_1(E_1^*(E)) W_2(E - E_1^*(E)) \, .
$$

To obtain the microcanonical temperature of the composite body of both systems 1 and 2, we focus on the second term – as the first (square root term) contains logs of the W's which will be small in the "thermodynamic limit". We will differentiate $\ln W(E)$, so the square root term will be a (small) additive factor, differentiating $\ln \ln W$, which we ignore.

$$
\begin{aligned}
\frac{\mathrm{d} \ln W(E)}{\mathrm{d}E} &= \frac{\mathrm{d} \ln W_1(E_1^*(E))}{\mathrm{d}E} + \frac{\mathrm{d} \ln W_2(E - E_1^*(E))}{\mathrm{d}E} + \quad \text{term from sq. root} \\
&\simeq \frac{W_1'(E_1^*(E))}{W_1(E_1^*(E))} \frac{\mathrm{d}E_1^*(E)}{\mathrm{d}E} + \frac{W_2'(E - E_1^*(E))}{W_2(E - E_1^*(E))} \left(1 - \frac{\mathrm{d}E_1^*(E)}{\mathrm{d}E}\right) \, .
\end{aligned}
$$

Now use Eq. (A.3) to rewrite this as:

$$
\begin{aligned}
\frac{1}{T^{\text{micro}}} = \frac{\mathrm{d} \ln W(E)}{\mathrm{d}E} &= \frac{1}{T_2^{\text{can}}} \left[\frac{\mathrm{d}E_1^*(E)}{\mathrm{d}E} + 1 - \frac{\mathrm{d}E_1^*(E)}{\mathrm{d}E}\right] \\
&= \frac{1}{T_2^{\text{can}}} = \frac{1}{T_1^{\text{can}}} \, .
\end{aligned}
$$

Thus the microcanonical temperature of the whole assembly is equal to the (identical) canonical temperatures of the two constituent bodies. (Up to corrections from the derivative of the log terms.)

B Zeroth Law of thermodynamics – non-examinable

The Zeroth law states if there are three subsystems are in thermal contact with each other, then if the first and second are in equilibrium and the second and third are also in equilibrium, then the first and third are as well. An example would be where, say, the second subsystem is a thermometer. If it is in equilibrium with the first and third, they must be at the same temperature – and hence in equilibrium with each other.

To derive this from the approach above, it is easiest to adopt a more symmetric treatment of the three subsystems than we used for two.

We need the integral representation of the delta function (from Y2 Mathematics, sec 3.4 example (vii)):

$$\delta(x-y) = \frac{1}{2\pi}\int \mathrm{d}\lambda\ \mathrm{e}^{-\mathrm{i}\lambda(x-y)}\ .$$

Then if the total number of configurations of the combined three subsystems, $W^{\text{tot}}(E)$, where E is the total energy, and the energies and numbers of configurations of the subsystems are E_i and $W_i(E_i)$, with $i = 1, 2, 3$, then:

$$\begin{aligned} W_{\text{tot}}(E) &= \int \mathrm{d}E_1\mathrm{d}E_2\mathrm{d}E_3\ W_1(E_1)W_2(E_2)W_3(E_3)\delta(E-[E_1+E_2+E_3]) \\ &= \frac{1}{2\pi}\int \mathrm{d}\lambda \int \mathrm{d}E_1\mathrm{d}E_2\mathrm{d}E_3\ W_1(E_1)W_2(E_2)W_3(E_3)\mathrm{e}^{-\mathrm{i}\lambda(E-[E_1+E_2+E_3])} \\ &= \frac{1}{2\pi}\int \mathrm{d}\lambda \int \mathrm{d}E_1\mathrm{d}E_2\mathrm{d}E_3\ \exp\left\{\ln W_1(E_1)+\ln W_1(E_1)+\ln W_1(E_1)-\mathrm{i}\lambda(E-[E_1+E_2+E_3])\right\}\ , \end{aligned}$$

Now we will find the maximum of the exponent, by taking partial derivatives with respect to E_1, E_2 and E_3, for example:

$$\frac{\partial}{\partial E_1}\left\{\ln W_1(E_1)+\ln W_1(E_1)+\ln W_1(E_1)-\mathrm{i}\lambda(E-[E_1+E_2+E_3])\right\} = \frac{\partial \ln W_1(E_1)}{\partial E_1}+\mathrm{i}\lambda = 0\ ,$$

for a stationary point. And similarly for the other two variables, i.e.

$$\frac{\partial \ln W_n(E_n)}{\partial E_n} = -\mathrm{i}\lambda \qquad \text{where } n = 1, 2, 3\ .$$

But we know that $k_{\text{B}} \ln W_n(E_n) = S_n$, the entropy of the nth subsystem, and that from thermodynamics we may identify (the subsystems have constant volumes, V_n):

$$\left(\frac{\partial S_n}{\partial E_n}\right)_{V_n} = \frac{1}{T_n} \quad \Rightarrow \quad \lambda = \frac{\mathrm{i}}{k_{\text{B}}T_n}\ ,$$

and so, since λ is a constant, $T_1 = T_2 = T_3$. I.e. all the subsystems are in equilibrium with each other.

In classical thermodynamics (see Y1 Temperature and Matter), the zeroth law predicts the existence of an equation of state relating the state variables, p and V to the temperature, T. The statistical mechanical derivation of the "equation of state" from the two subsystem version (in main body of section) is a little subtle (presently it will become easier by considering the Free energy).

One has

$$\left(\frac{\partial \ln W_n(E_n, V_n)}{\partial E_n}\right)_{V_n} = \frac{1}{k_{\rm B}T} \quad \text{and} \quad \left(\frac{\partial \ln W_n(E_n, V_n)}{\partial V_n}\right)_{E_n} = \frac{p}{k_{\rm B}T} \ ,$$

the so-called "caloric" and "mechanical" equations of state. They may be turned into "the" equation of state by solving one for E_n and substituting into the other to leave a function depending on p, V and T.

With the Helmholtz free energy, $F(T, V)$, the relations are

$$\left(\frac{\partial F}{\partial V}\right)_T = -p(V, T) \quad \text{and} \quad \left(\frac{\partial F}{\partial T}\right)_V = -S(V, T) \ .$$

The lefthand relation is immediately "the" equation of state involving $p(V, T)$, V and T – with the left hand side depending on T and V. The "other", righthand, equation of state provides $S(T, V)$.

This distinction is manifested for an ideal gas in the statements that one needs to specify *both* $pV = nRT$ *and* that $U(T)$ alone, not a function of V as well.